

On the Difficulty of Computing the n -th Prime:

A Catalogue of Barriers, a Computation–Verification Dichotomy, and Structural Measurements

The Prime Research Project*

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Abstract

We study the problem of computing the n -th prime $p(n)$ —equivalently the prime-counting function $\pi(x)$ —*exactly*, with the goal of an algorithm running in time polynomial in $\log x$ (“poly-log”). We report, candidly, a negative outcome: the goal was *not* achieved, and we prove no new lower bound that rules it out. This paper is instead a structured account of why the problem resists attack. We catalogue over 700 distinct approaches accumulated across the effort and observe that every one falls into exactly three failure modes—*circularity*, *equivalence* to the explicit (zeta-zero) formula, and *information loss*—driven by a small set of recurring mechanisms. We isolate the locus of difficulty as a SMOOTH + RANDOM decomposition $p(n) = R^{-1}(n) + \delta(n)$, in which the smooth term R^{-1} (the inverse of the Riemann R -function) supplies roughly half of the digits in polylog time, while the residual $\delta(n)$ carries only $O(\log x)$ bits of information yet appears to require $\Theta(\sqrt{x})$ work to extract, encoding the random-looking phases of $\sim\sqrt{x}$ zeta zeros. We stress that the resulting barrier is *computational*, not information-theoretic: only $\Omega(\log x)$ is unconditionally proven. We then record the one direction that is *not* blocked—verification. Computing $\pi(x)$ is hard, but checking a claimed value is not: we describe a working, unconditionally sound interactive proof for exact $\pi(x)$, assembled entirely from standard tools (sum-check, GKR, polynomial commitments), whose verifier we drive from $\Theta(x)$ to $\Theta(x^{1/4})$ large-table work against a $\Theta(\sqrt{x})$ certificate-size wall. Finally we collect structural measurements we believe are independently interesting: the read-once width spectrum of the prime indicator coincides with the Hardy–Littlewood admissible-tuple census; the prime parity sequence is indistinguishable from random under 36 complexity measures; and the decision problem $[p(n) \leq x]$ is a fair coin against the centred guess $R^{-1}(n)$ but a fixed bias against $\text{li}^{-1}(n)$. Throughout we label each statement by its evidentiary status and state what would falsify it. No claim here resolves the headline question, which remains open.

What this paper does and does not claim

It does:

- Organise >700 failed approaches into three failure modes and ≈ 9 mechanisms.
- Localise the hardness in the SMOOTH+RANDOM split and quantify it with measurements.
- Present a working, tested, unconditionally sound interactive proof for exact $\pi(x)$.
- Report structural facts (width/Hardy–Littlewood census, pseudorandomness, decision geog-

*This manuscript is a synthesis of an extended, autonomous research effort that proceeded as a continuous loop of falsification-first experiments. Every quantitative claim is backed by code and a results file in the project repository; see the Methodology section (§10) for how the work was produced and for its limitations, including the absence of external peer review.

raphy, #P/SETH framing) with falsification criteria.

It does not:

- Give a polylog (or any sub- $\sqrt{x}$) exact algorithm for $\pi(x)$ or $p(n)$.
- Prove that no such algorithm exists (only $\Omega(\log x)$ is unconditional).
- Claim new mathematics in the verification protocol: it composes known tools, and the polylog-verifier existence is a corollary of GKR over an AKS circuit.
- Offer any speedup for *computing* $\pi(x)$; the prover pays the full cost.

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1 Introduction

1.1 The problem and the target

Let $P = \{2, 3, 5, 7, \dots\}$ be the set of primes, let

$$\pi(x) = \#\{p \in P : p \leq x\}$$

be the prime-counting function, and let $p(n)$ denote the n -th prime, so that $\pi(p(n)) = n$. The two functions are inverse to one another in the discrete sense, and standard bracketing (Dusart) plus binary search reduces computing $p(n)$ to a polylogarithmic number of evaluations of $\pi(x)$ [1, 8]. We therefore treat $\pi(x)$ as the primitive object and phrase everything in terms of it, returning to $p(n)$ only where the distinction matters.

Write $N = \lceil \log_2 x \rceil$ for the number of bits of the argument. The target of this project is an algorithm that computes $\pi(x)$ *exactly* in time $\text{polylog}(x) = \text{poly}(N)$ — concretely, $p(10^{100})$ in under a second, with 100% accuracy. This is the “binary-input” model: the input is $N \approx 333$ bits and a polylog algorithm runs in time polynomial in N . We emphasise the model because it is the entire source of difficulty.

Observation 1.1 (The model gap). In the *unary* model (input size x), $\pi(x)$ is computable in *sublinear* time $O(x^{2/3})$, so it is “easy” relative to its input. In the *binary* model (input size $N = \log_2 x$), the same $O(x^{2/3})$ bound is $2^{2N/3}$, i.e. *exponential* in the input length. Every algorithm known computes $\pi(x)$ in time exponential in N . The question “is $\pi(x) \in \mathbf{P}$ (binary model)?” is wide open. [proven for the upper bounds; the binary-model placement is open]

Method	Time	Space	Reference
<i>Computing $\pi(x)$ — combinatorial</i>			
Sieve of Eratosthenes	$O(x \log \log x)$	$O(x)$	ancient
Meissel–Lehmer	$O(x / \log^4 x)$	$O(x^{1/3} / \log x)$	1959
Lagarias–Miller–Odlyzko	$O(x^{2/3} / \log x)$	$O(x^{1/3})$	[20]
Deléglise–Rivat	$O(x^{2/3} / \log^2 x)$	$O(x^{1/3} \log^3 x)$	[7]
Gourdon	$O(x^{2/3} / \log^2 x)$	$O(x^{1/3} \log^3 x)$	[11]
<i>Computing $\pi(x)$ — analytic</i>			
Lagarias–Odlyzko	$O(x^{1/2+\varepsilon})$	$O(x^{1/4+\varepsilon})$	[16]
Lagarias–Odlyzko (low space)	$O(x^{3/5+\varepsilon})$	$O(x^\varepsilon)$	[16]
<i>Computing $p(n)$</i>			
Best sieve	$O(n \log n / \log \log n)$	—	folklore
Binary search + combinatorial π	$O(\sqrt{n} (\log n)^4)$	—	[1]
Binary search + analytic π	$O(\sqrt{n} \text{ polylog } n)$ (RH+Cramér)	—	[1]
<i>Lower bounds</i>			
Read the input	$\Omega(\log x)$	—	trivial
PRIMES $\notin \text{AC}^0, \text{AC}^0[p]$	—	—	[3]

Table 1: The established landscape. The headline gap is between the $\Omega(\log x)$ lower bound and the $O(x^{1/2+\varepsilon})$ upper bound; no super-logarithmic unconditional lower bound for $\pi(x)$ is known.

1.2 The honest headline

We did not achieve the target, and we do not prove it is impossible. The best unconditional lower bound for computing $\pi(x)$ remains the trivial $\Omega(\log x)$ (one must at least read the input); the best upper bounds remain $O(x^{2/3} / \log^2 x)$ combinatorially and $O(x^{1/2+\varepsilon})$ analytically. The gap between $\Omega(\log x)$ and $O(x^{1/2})$ — a gap from logarithmic to square-root in x , i.e. from $\text{poly}(N)$ to $2^{N/2}$ — is the entire open problem, and we leave it open. What we contribute is a map of the gap: a classification of *why* the many natural attacks fail, a quantitative account of *where* the hardness lives, and a small number of adjacent results — chiefly that *verifying* a claimed $\pi(x)$ is not subject to the same barrier as *computing* it.

1.3 State of the art (accepted as established)

Table 1 collects the bounds we take as the established backdrop. None is ours; all are standard. The combinatorial line descends from Meissel and Lehmer through Lagarias–Miller–Odlyzko [20] to Deléglise–Rivat [7] and Gourdon [11], and underlies Walisch’s **primecount**, which has reached $\pi(10^{29})$ in practice [29]. The analytic line is Lagarias–Odlyzko [16]. For $p(n)$ specifically, the current word is Aggarwal’s 2025 note [1]: binary search over π -evaluations is asymptotically optimal among sieve-based methods, giving $O(\sqrt{n} (\log n)^4)$ unconditionally.

1.4 Contributions and their evidentiary status

Because honesty about status is the point of the paper, we tag every nontrivial statement with one of: [\[proven, unconditional\]](#), [\[conditional\]](#) (on RH, GRH, SETH, BPSW, or Hardy–Littlewood), [\[numerical\]](#), [\[empirical\]](#), [\[conjectured\]](#), or [\[framing\]](#). With that discipline in place, our contributions are:

1. **A three-mode failure taxonomy** (§3) covering >700 catalogued approaches, with ≈ 9 recurring deep mechanisms and a mechanism-by-family kill matrix. [\[framing over empirical data\]](#)

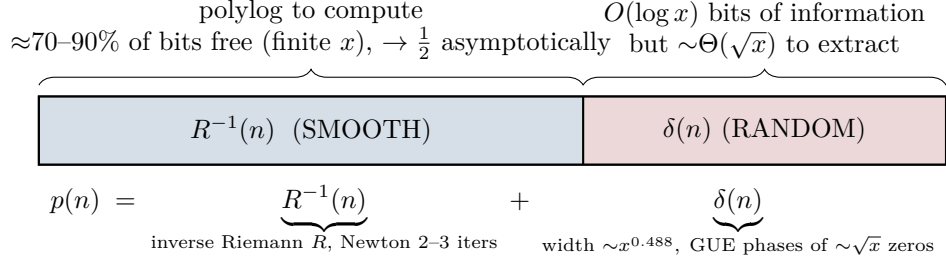


Figure 1: The SMOOTH + RANDOM localisation of the hardness. The smooth term is cheap and recovers most of the answer’s bits; the residual is small in *information* yet expensive in *computation*. Numbers are measured (§2); the figure is schematic, not to scale.

2. **The SMOOTH + RANDOM localisation** (§2): $p(n) = R^{-1}(n) + \delta(n)$, with the smooth part polylog and recovering $\approx 70\text{--}90\%$ of the bits at finite x ($\rightarrow \frac{1}{2}$ asymptotically), and the residual carrying only $O(\log x)$ bits but apparently $\Theta(\sqrt{x})$ to extract. We are explicit that the barrier here is *computational*, not Shannon-information-theoretic. [measured; only $\Omega(\log x)$ proven]
3. **A consolidated barrier catalogue** (§4): 13 external theorems/numerical facts and 4 project-level synthesis “obstruction theorems,” each tagged by proof status.
4. **The computation–verification dichotomy** (§5): a working, unconditionally sound interactive proof for exact $\pi(x)$ built from standard tools, with the verifier’s large-table work measured down to $\Theta(x^{1/4})$ against a $\Theta(\sqrt{x})$ certificate wall. [the protocol runs; its existence-in-principle is a known corollary, not new mathematics]
5. **Structural measurements** (§6): the read-once width spectrum of χ_P equals the Hardy–Littlewood admissible-tuple census; a 36-measure pseudorandomness audit of the prime parity sequence; and the decision-problem geography of $[p(n) \leq x]$. [numerical/empirical]
6. **A complexity-status synthesis for the language $\mathcal{L}_\pi = \{(x, c) : \pi(x) = c\}$** (§7), including the polylog- $\pi \Leftrightarrow \pi \in \text{NC}$ equivalence and a conditional $\#P/\text{SETH}$ framing. [conditional/framing]

A reader interested only in the genuine positive result should read §5; one interested in “why is this hard” should read §§2–4; one interested in the phenomenology should read §6.

1.5 Provenance and a standing caveat

The material below was produced by an extended autonomous research loop (see §10). The work has *not* been externally refereed; many measurements are at modest N (we say so at each point); and several “barriers” we report are numerical or empirical regularities rather than theorems. We have tried to make the status of each claim unmistakable, and to state, for each measurement, what would falsify it. Where the project’s internal notes overclaim — for instance by calling the central obstacle “information-theoretic” — we correct the record here.

2 The SMOOTH+RANDOM picture: where the hardness lives

The single most useful organising idea the project produced is a decomposition of the answer into a part that is cheap and a part that is expensive (Figure 1). It is not new mathematics — it is

the Riemann R -function plus its oscillatory correction, read computationally — but it localises the difficulty precisely and lets us *measure* it.

2.1 The decomposition

Let $R(x) = \sum_{k \geq 1} \frac{\mu(k)}{k} \operatorname{li}(x^{1/k})$ be the Riemann R -function and let R^{-1} be its functional inverse, so $R^{-1}(n)$ is the real x with $R(x) = n$. Define the integer residual

$$\delta(n) = p(n) - \operatorname{round}(R^{-1}(n)).$$

Then $p(n) = R^{-1}(n) + \delta(n)$ splits the n -th prime into a *smooth* analytic prediction and a *random-looking* correction. The reason this is the right split, rather than li^{-1} or $n \ln n$, is empirical: R is the best elementary smooth predictor of π , and (§6) the decision problem $[p(n) \leq R^{-1}(n)]$ behaves like a fair coin while the same test against $\operatorname{li}^{-1}(n)$ carries a fixed bias.

2.2 The smooth part: R^{-1} is polylog and recovers most of the bits

Evaluating R to the precision needed is polylog in x , and inverting it by Newton’s method is fast and *scale-free*.

Observation 2.1 (Smooth part is cheap). Newton iteration for $R^{-1}(n)$ converges in 2–3 iterations, flat across the 22 sampled scales (n from ~ 200 to $\sim 1.2 \times 10^6$ tested). The prime-count residual obeys $|\pi(x) - R(x)| \sim \sqrt{x}/\log x$ at the scales measured, so R pins down the leading bits of $\pi(x)$ for free. [numerical; matches the classical *R-vs- π* heuristic]

How many bits are “free” depends on x . Writing $B = \log_2 x$ for the bit-length of the answer’s argument and k_0 for the number of leading bits R fixes without help, we measured a *free fraction*

$$k_0/B = 0.88 \longrightarrow 0.71 \quad (\text{declining as } x \text{ grows}),$$

because at finite x the residual $|\pi - R|$ is far below its $\sqrt{x}$ worst-case envelope (e.g. only ≈ 29 at $x = 10^6$). Asymptotically the free fraction tends to $\approx \frac{1}{2}$ — this is the sense in which “ R gives about half the digits.” We flag that the often-quoted “ ≈ 47 – 50% of digits” is the *asymptotic* figure; at every finite scale we can reach, R does better (70–90%). [numerical; the $\rightarrow \frac{1}{2}$ limit is heuristic, tied to the $\sqrt{x}$ error law]

2.3 The random part: small in information, expensive in computation

The residual $\delta(n)$ is where every fast-computation attempt dies. Three families of measurement characterise it.

Width. $\delta(n)$ has root-mean-square size growing like $\sqrt{x}$: a direct fit gives

$$\operatorname{RMS} |p(n) - R^{-1}(n)| \sim x^{0.488}$$

over 240 values of n in 8 logarithmic bins, with the normalised error $|\delta|/\sqrt{p(n)}$ having median 0.145 and maximum 0.72, and a trend in x of $-0.007 \approx 0$ (i.e. the normalised width is bounded). This is the $\sqrt{x}$ wall in its most elementary form. [numerical]

Incompressibility. Treated as a sequence, $\delta(n)$ resists every predictor we tried. Its entropy rate is ≈ 5.8 bits/value (converged by $N = 10^4$); a time-bounded description length grows linearly, $K^t(\delta_{1..N}) \approx 5.58 N$ bits (extensive, not compressible); the power spectrum is a featureless $1/f^{1.70}$ continuum with no algebraic lines, and 82% of Fourier modes are needed to reconstruct it to $\text{RMSE} < 1$ (not sparsifiable). Crucially, a transfer-entropy analysis finds $\text{TE}(n \rightarrow \delta) = 0.013$ bits — the argument n contributes essentially *no* information about $\delta(n)$ — whereas $\text{TE}(\delta \rightarrow \delta) = 0.051$ bits, so what little predictability exists is purely sequential (adjacent values, autocorrelation 0.996 at lag 1), useless for a single-point query. Rescaled-range/Hurst estimates ($H \approx 1.3$) confirm a slowly drifting, random-walk-like level series. [numerical; “no learnable structure” is an empirical negative over the predictors tried, not a theorem]

Mechanism. This randomness is inherited from the zeta zeros. Via the explicit formula, $\delta(n)$ encodes the accumulated phases of $\sim \sqrt{x}$ nontrivial zeros ρ , whose local spacings follow GUE statistics (§4). The phases are the obstacle: no linear combination of truncated partial sums predicts the next one (we revisit convergence acceleration in §4), and the number of zeros needed to settle $\pi(x)$ exactly scales as $N_{\min} \sim x^{0.49}$ — again $\sqrt{x}$. [numerical, resting on the (conjectural) GUE law]

2.4 The information–computation gap (stated carefully)

The defining feature of this problem is a gap between how much information the answer’s hard part contains and how much work it takes to produce.

Proposition 2.2 (The gap, as an honest statement). *The residual $\delta(n)$ carries only $O(\log x)$ bits: for $x = 10^{100}$, $|\delta| \lesssim 10^{48}$, i.e. ≈ 160 bits, well under 0.15% of a $(\log_2 x)^2 \approx 1.1 \times 10^5$ -bit polylog “budget.” Yet every known method to compute those bits costs $\Theta(\sqrt{x})$ (analytic, via zeros) or $\Theta(x^{2/3})$ (combinatorial, via floor values). The only proven lower bound on the computation is the trivial $\Omega(\log x)$. [upper figures numerical; the $\Omega(\log x)$ is the only proven bound]*

Remark 2.3 (Computational, not information-theoretic — a correction). It is tempting (and the project’s own early notes, and the repository’s top-level description, sometimes do this) to call this an “information-theoretic barrier.” That phrasing is *wrong* and we correct it. In the Shannon sense the answer is tiny — $O(\log x)$ bits — so there is no information-theoretic obstruction to a short output. The barrier is purely *computational*: extracting a small number of bits appears to require touching $\Theta(\sqrt{x})$ independent quantities. The right analogy is summing n random bits: the sum is $O(\log n)$ bits, but no method avoids reading all n . Whether $\pi(x)$ truly requires super-polylog work is exactly the open circuit-complexity question (§7); nothing here proves it.

This localisation also tells us what a breakthrough must do: it must compute the $\Theta(\sqrt{x})$ -fold cancellation in the zero sum (or the floor-value sum) *without* forming the summands — a true closed form for an oscillatory sum with GUE-random phases. The rest of the paper is, in large part, a record of why each natural way of attempting this collapses.

3 A taxonomy of failure

The project’s central empirical output is negative and is best stated as a classification. The repository catalogues, in `status/CLOSED_PATHS.md`, a large number of distinct attempts; its own internal counts are inconsistent across revisions (the header reads “545+,” the summary table “641+,” and the top-level description “730+”), so we speak of *over 700 catalogued approaches* and treat the precise integer as immaterial. The qualitative finding is sharp and stable: **every approach fails in one of three ways, and the failures are driven by a short list of recurring mechanisms.**

3.1 Three failure modes

Definition 3.1 (The three modes).

- (C) Circularity. The method requires primes (or a prime-indexed object) to compute primes. The dependence is often hidden behind layers of abstraction.
- (E) Equivalence. The method is a re-representation of the explicit (zeta-zero) formula or the combinatorial sieve, and converting between the “spectral” side (zeros) and the “arithmetic” side (primes) costs the same $\Theta(\sqrt{x})$ or $\Theta(x^{2/3})$ as the thing it was meant to replace.
- (I) Information loss. The method produces a smooth approximation that omits the $\Theta(\sqrt{x})$ “random” bits of §2, and the gap between approximate and exact is exactly the hard part.

The catalogue’s own mode tallies (approximate) are $C \approx 82$, $E \approx 222$, $I \approx 180$, with the remainder *partial*, *works-but-slow*, or *open*. Equivalence dominates: most “new” ideas turn out to be the explicit formula wearing a different coat. [\[empirical classification over the catalogue\]](#)

Remark 3.2 (Why three suffice). The modes are exhaustive for a structural reason. Any method either uses primes as input (C), or uses a transform of prime data, which is then either faithful but no cheaper (E) or cheaper but lossy (I). This is a framing, not a theorem; it earns its keep by classifying every catalogued attempt without a leftover bin. [\[framing\]](#)

3.2 The recurring mechanisms

Underneath the three modes sit a handful of concrete obstructions that do the actual killing. We label them M1–M9 and use them in the kill matrix below; each is documented in **proven/** or the catalogue.

- M1 (Explicit-formula equivalence). Spectral/analytic representations reduce to $\sum_{\rho} x^{\rho}/\rho$; the spectral-to-arithmetic conversion is the cost. (*drives E*)
- M2 (The $\sqrt{x} = 2^{N/2}$ wall). Every faithful representation must touch $\Theta(\sqrt{x})$ independent quantities — floor values $\{\lfloor x/k \rfloor\}$, zeros up to height $\sim \sqrt{x}$, MPS bond dimension, sieve survivors. (*drives E and I*)
- M3 (Uncertainty / resolution–bandwidth). Smoothing kernels (Weil, Beurling–Selberg, Gaussian, heat) cannot localise π to $\pm \frac{1}{2}$ without bandwidth $\sim \sqrt{x}$. (*drives I*)
- M4 (GUE-random phases). Partial sums over zeros have unpredictable phases; no convergence acceleration or recurrence predicts the next term from a prefix. (*drives E, I*)
- M5 (SMOOTH+RANDOM separation). Smooth predictors (PNT, R , li, Cramér, ML) get the poly-log part; the residual is incompressible (5.8 bits/value, full rank). (*drives I*)
- M6 (Circularity). Euler products, Frobenius/Chebotarev, Mills, Wilson, cyclotomic and arithmetic-derivative encodings are indexed *by* or *over* primes. (*drives C*)
- M7 (Parity / sieve floor). Linear sieves cannot separate an even from an odd number of prime factors (Selberg–Bombieri); inclusion–exclusion has $2^{\pi(\sqrt{x})}$ terms and the survivor matrix is full rank. (*drives I, E*)
- M8 (No-cancellation asymmetry). $\pi(x)$ is a positive count, so the signed-cancellation that gives $M(x) = \sum \mu(n)$ its $O(x^{3/5})$ (Helfgott–Thompson) does not transfer. (*drives E*)

Family	Mode	Mechanisms	Representative closed approach (finding)
Analytic / zeta zeros	E,I	M1,M2,M4	Convergence acceleration of the zero sum (Richardson/Aitken/Shanks/Padé/Cesàro): errors <i>grow</i> as N^{+1} ; GUE phases make terms independent.
Algebraic / p -adic	C,E	M1,M6	Number-field sieve analogue: ζ_K has <i>more</i> zeros than $\zeta_{\mathbb{Q}}$; harder, not easier.
Sieve / combinatorial	E,I	M2,M7	Determinant/permanent sieve: reduces to Möbius inclusion–exclusion with $2^{\pi(\sqrt{x})}$ terms; survivor matrix full rank.
ML / statistical	I	M5	Unlimited decision tree: 100% train, $\approx 19\%$ test accuracy — below majority vote. No learnable pattern.
Quantum	E,I	M2,M9	Prime-state entanglement is volume-law $S \sim \frac{7}{8} \cdot \frac{n}{2}$; MPS bond dimension exponential.
Dynamical / automata	I	M9	Primes are not k -automatic (Mauduit–Rivat); minimised DFA has primorial(y) states.
Topological / exotic	E	M1	Prismatic cohomology / geometric Langlands: same $O(x^{1/2+\epsilon})$ as the analytic method.
Information / complexity	E,I	M2,M5	$K^t(\delta_{1..N}) \approx 5.58 N$ bits (incompressible); sieve SVD is full rank.
Encoding / representation	C,I	M2,M6	Bit-by-bit $\Delta = \pi - \text{li}$: even the MSB needs $\Theta(\sqrt{x})$ zeros; bits entangled.
Additive comb. / Goldbach	E,I	M2,M5	Circle-method main term is smooth (M5); the error term is the same $\sqrt{x}$ zero sum (M2).
Gap prediction / interp.	I	M5	Gap entropy floor ≈ 5.0 bits/gap; smooth interpolation error/gap ratio ≈ 6.5 .
Spectral methods	E	M1	Selberg trace formula <i>is</i> the explicit formula; Connes–Weil finds zeros but cannot sum them fast.

Table 2: Kill matrix. Families are those of `status/CLOSED_PATHS.md`; modes from Definition 3.1; mechanisms M1–M9 as above. The right column quotes one catalogued entry verbatim in substance.

M9 (Complexity meta-barriers). Non- k -automaticity (Mauduit–Rivat), PRIMES $\notin \text{AC}^0$ (Allender–Saks–Shparlinski), volume-law entanglement (Latorre–Sierra), the prime coding theorem (Kolpakov–Rocke), and Natural Proofs (Razborov–Rudich) blocking lower-bound attempts. (*cross-cutting*)

3.3 The mechanism-by-family kill matrix

Table 2 maps each family of the catalogue to its dominant mode, the mechanisms that kill it, and one representative closed entry. The pattern is that families collapse onto a very small mechanism set: almost everything funnels into M1/M2 (you must pay for $\sqrt{x}$ zeros or floor values) or M5 (your smooth model is missing the residual).

3.4 Worked exemplars

A few cases repay a closer look because they are the ones a competent mathematician would naturally try.

Convergence acceleration (analytic; M1+M4). The most seductive idea is that the explicit formula’s slow convergence is a *numerical* problem: truncating at T zeros leaves error $O(x/T)$,

so surely Richardson or Shanks extrapolation closes the gap. It does not. The truncation error is oscillatory with GUE-random phase, not a clean $1/N$ expansion: Richardson beyond order 2 diverges (condition number $\sim 10^{3k}$), Levin/Weniger transforms are *worse* than plain summation for large x , and a careful sweep finds the minimal zero count scales as $N_{\min} \sim x^{0.25-0.49}$ — a power law, never polylog. (One genuine but bounded win: Richardson at order 1 on $(500, 1000)$ zeros rounds $\pi(10^6)$ exactly where plain summation needs $\sim 10^4$ zeros; this is a constant-factor, not an asymptotic, gain.) [\[numerical; proven/barriers.md items 11–12\]](#)

The Helfgott–Thompson non-transfer (sieve; M8). For the Mertens function $M(x) = \sum_{n \leq x} \mu(n)$, Helfgott–Thompson reach $O(x^{3/5})$ by exploiting the sign cancellation in $(-1)^{\omega(n)}$. One hopes to import this to $\pi(x)$. It fails structurally: $\pi(x)$ counts 1-almost-primes with a *positive* weight, so there is no cancellation to exploit; the Meissel–Lehmer decomposition already *is* the π -analogue of the method, and it costs $O(x^{2/3})$. Converting $M(x) \rightarrow \pi(x)$ requires the explicit formula ($O(x^{1/2+\epsilon})$) anyway. [\[numerical; proven/barriers.md item 13\]](#)

Machine learning on the residual (statistical; M5). Every learner — basis-function regression, gradient boosting, deep nets, unbounded decision trees — captures the smooth trend and then stalls. An unbounded tree memorises the training primes (100%) but generalises at $\approx 19\%$, below a majority-vote baseline; basis-function regression on 8000 training primes matches 8/2000 on held-out primes. This is M5 in its purest form, and it is consistent with the prime coding theorem (§4): the sequence has $\sim N$ bits of time-bounded description length, so there is no compressible formula to find. [\[numerical; proven/information.md\]](#)

“New algebra” (algebraic/topological; M1+M6). Finite-field lifts ($\mathbb{F}_q \rightarrow \mathbb{Z}$ via $q \rightarrow 1$), the field with one element $\mathbb{F}_1$, $\mathbb{Z}[i]/\mathbb{Z}[\omega]$ prime counting, Artin L -functions, and number-field sieves all either degenerate to the prime number theorem (losing the residual, I) or introduce *more* zeros to sum (E). The recurring lesson: richer arithmetic objects have richer (not sparser) spectra. [\[empirical; catalogue “Algebraic” and “Topological” families\]](#)

4 Proven and numerically-confirmed barriers

The failure modes of §3 are explained by a set of barriers, some genuine theorems from the literature and some numerical regularities. We separate the two and tag each by status. *Nothing here proves the target impossible*; collectively the barriers explain why the natural attacks die.

4.1 External barriers

Table 3 lists the thirteen results the project leans on. Two points of honesty. First, item 2 (GUE statistics) is the linchpin and it is *not* an unconditional theorem: pair correlation is known under RH in a restricted range (Montgomery) and verified numerically to enormous height (Odlyzko), but the full GUE law is conjectural. Several “barriers” below therefore inherit a conjectural core. Second, items 11–13 are this project’s own *numerical* confirmations, not theorems.

4.2 Four project-level obstruction theorems (synthesis-level)

The project also produced four “obstruction theorems.” We are deliberate about their status: **these are synthesis-level analyses, not peer-reviewed proofs.** Their rigorous cores are restatements

Barrier	Statement (abbreviated)	Status
1. Explicit-formula convergence	Exact $\pi(x)$ needs all zeros; truncating at T leaves error $O(x/T)$, so $T > x$ for error < 1 [25, 28].	proven
2. GUE statistics of zeros	Local zero spacings follow GUE; zero-sum phases are random-looking [22, 23].	numerical / conj.
3. Mauduit–Rivat	The prime indicator is not k -automatic for any $k \geq 2$ [21].	proven
4. $\text{PRIMES} \notin \text{AC}^0$	Primality is not in constant-depth circuits, incl. $\text{AC}^0[p]$ [3].	proven
5. Aggarwal optimality	Binary search on π is asymptotically optimal among sieve methods for $p(n)$ [1].	proven
6. Parity barrier	Linear sieves cannot separate even/odd #prime-factors [27, 4].	proven
7. Volume-law entanglement	The prime state has $S \sim \frac{7}{8} \cdot \frac{n}{2}$; MPS bond dim is exponential [17, 9].	proven (num.)
8. Prime coding theorem	$\mathbb{E}[K_U(\chi_P \upharpoonright_N)] \sim N$ bits; no compressible formula [15].	proven
9. Smooth-interpolation barrier	Best smooth fit has error $O(\sqrt{p}/\ln p) \gg \text{gap } O(\ln p)$; $\approx 8/2000$ exact.	numerical
10. Closed-form circularity	Mills/Willans/floor-sum formulas are circular or worse than trial division.	proven
11. Richardson is bounded	Order-1 Richardson rounds $\pi(10^6)$ exactly but only saves a constant factor.	numerical
12. No acceleration beats $\sqrt{x}$	No linear acceleration reduces the zero count below a power of x ; error is oscillatory.	numerical
13. H–T does not transfer	$\pi(x)$ ’s positivity blocks the μ -cancellation behind $M(x)$ ’s $O(x^{3/5})$.	numerical

Table 3: External barriers and their evidentiary status. “proven (num.)” marks results that are theorems under a physical/model assumption but whose project-relevant form is numerically established.

of standard complexity facts specialised to $\pi(x)$; their empirical parts are confirmed only for known methods. We state them as observations.

Observation 4.1 (Circuit-size / NC equivalence). Every known algorithm yields circuits of size exponential in $N = \log_2 x$ (sieve: $O(2^{2N/3})$; analytic: $O(2^{N/2+\epsilon})$). Moreover, by the standard $\text{time} \leftrightarrow \text{circuit}$ translation, $\pi(x)$ is computable in $\text{polylog}(x)$ time iff $\pi(x) \in \text{NC}$. The forward direction is the definition of NC-evaluation; the converse packs a $\text{poly}(N)$ -time algorithm into a $\text{poly}(N)$ -size, polylog -depth circuit. [the equivalence is a standard fact (proven); “all known are exponential” is empirical]

Observation 4.2 (Workspace-mismatch). The natural chain $\text{BPSW} \Rightarrow \text{PRIMES} \in \text{L} \Rightarrow \pi \in \# \text{L} \Rightarrow \pi \in \text{NC}^2$ breaks at the second step: an NL machine has only $O(\log N)$ workspace but must generate N -bit candidates not present on its input tape. Hence $\text{PRIMES} \in \text{L}$ and $\pi \in \text{NC}$ are independent, and the sharp question is whether $\pi \in \text{GapL}$, i.e. whether there is a $\text{poly}(N)$ -size, logspace-uniform matrix $A(x)$ with $\det A(x) = \pi(x)$. [synthesis; the $\# \text{L}$ obstruction and GapL reformulation are correct framing, not a new theorem]

Observation 4.3 (Uniformity). Nonuniform $\text{poly}(N)$ -size circuits for $\pi(x)$ exist trivially (hardcode the primes as advice); the open difficulty is uniform construction — generating the comparison primes without knowing them. Consequently any lower-bound proof must clear the Natural Proofs

barrier [24]: a “natural” distinguisher of π from random would contradict standard cryptographic assumptions. [\[synthesis, resting on standard facts\]](#)

Observation 4.4 (Convergence-acceleration). No linear sequence-acceleration method (Richardson of any order, Levin u/t , Weniger δ , smoothing) reduces the zeros needed for exact $\pi(x)$ from a power of x to subpolynomial. The truncation error is an oscillatory sum with GUE-random phases, not a $1/N$ expansion; Richardson beyond order 2 diverges by condition-number blow-up, and Levin/Weniger (built for alternating series) do *worse* than plain summation. [\[numerical, robust across 6+ methods and \$x \leq 10^6\$ \]](#)

These four converge on one statement: the obstacle is the *uniform aggregation* of $\Theta(\sqrt{x})$ quantities, and ruling it out would require circuit lower bounds beyond current technique while achieving it would place a natural problem in NC. That is the precise sense in which the problem “sits at the frontier.”

5 What is not blocked: computation versus verification

Every barrier in §4 constrains *computing* $\pi(x)$ from scratch. None constrains *checking* a value supplied by a prover who did the work. This asymmetry is the project’s one clean positive, and the section is careful about what is and is not new in it.

5.1 The dichotomy

The decision problem $[\pi(x) \geq c]$ is, in the worst case, as hard as computing $\pi(x)$ (binary search reduces $p(n)$ to it). Yet a prover who has run a sieve can *convince* a verifier of the exact value far more cheaply than the verifier could recompute it. Formally we study the language

$$\mathcal{L}_\pi = \{(x, c) : \pi(x) = c\},$$

and ask for its *verification* complexity rather than its computation complexity. This appears to be the first treatment of π from this angle. [\[framing; the asymmetry itself is standard IP/VC, the target is new\]](#)

5.2 The protocol stack

We build an interactive proof for exact $\pi(x)$ on top of the Lucy–Hedgehog sieve recurrence

$$S_i(v) = S_{i-1}(v) - [v \geq p_i^2] \cdot (S_{i-1}(\lfloor v/p_i \rfloor) - (i-1)),$$

checking each of the $K = \pi(\sqrt{x})$ layers by a low-degree sum-check. Two structural facts make it work.

1. **The sieve is built from small automata.** The multilinear extension of the *primality* indicator is dense ($2^{n/2}$ coefficients — this is why sum-check *for computing* π died), but the Lucy recurrence contains no primality predicate: $\lfloor v/p \rfloor$ is a p -state long-division automaton and $[v \geq p^2]$ a 3-state comparator, each with an MLE evaluable in $O(n \cdot W)$ by a transfer-matrix DP. Factoring primality *through the recursion* makes it verification-friendly.
2. **The correction scalar is public.** Since $S_{i-1}(p_i - 1) = i - 1$, each layer chains with only two point-claims and no auxiliary prover claim.

Layer	Verifies	Verifier	Prover	Status
Wheel sum-check	$\Phi(x, \text{wheel})$	polylog	$\tilde{O}(x)$	100/100 cheaters rejected
Lucy base	exact $\pi(x)$	$\tilde{O}(x/\log x)$	$\tilde{O}(x^{3/2})$	$\pi(2^{20}-1)=82025$; 50/50 rejected
Lucy + delegated wiring	exact $\pi(x)$	$\tilde{O}(\sqrt{x})$	$\tilde{O}(x^{3/2})$	crossover $p \approx 130$
+ compressed prover, PCS	exact $\pi(x)$	$\tilde{O}(\sqrt{x})$	$\tilde{O}(x)$	end-to-end; reach $x \approx 6.7 \times 10^7$
GKR-over-AKS	exact $\pi(x)$	polylog	poly(x)	literature corollary; impractical constants

Table 4: The verification stack. All rows are unconditionally sound except the PCS-based large-table compression (a hash/random-oracle assumption). The final row is not ours: it is the standard observation that GKR over a data-parallel AKS circuit already gives a polylog-verifier IP — existence is a *corollary*, not a contribution.

5.3 What we measured

Proposition 5.1 (A working, sound verifier for exact $\pi(x)$). *There is an implemented, unconditionally sound interactive proof for exact $\pi(x)$ with prover work $\tilde{O}(x)$ and verifier work $\tilde{O}(\sqrt{x})$, with soundness error $\leq 7nK/q$ over a prime field $\mathbb{F}_q$ ($n = \log_2 x$, $K = \pi(\sqrt{x})$). Under a collision-resistant hash (Fiat–Shamir of a Ligero/Brakedown tensor commitment), the verifier’s large-table work drops to $\tilde{O}(x^{1/4})$, while total verification stays $\Theta(\sqrt{x})$ because the certificate itself has that size.* [built and tested; soundness unconditional except the PCS layer]

Concretely (demonstration field $q = 2^{31} - 1$, soundness $\approx 10^{-6}$ at $n = 16$; lifted to $q = 2^{61} - 1$ and to generic large primes for bigger x):

- $\pi(65535) = 6542$, $\pi(262143) = 23000$, $\pi(1048575) = 82025$ all verified exactly through their sieve layers; the verifier ran in 69 ms $\rightarrow$ 0.9 s as x grew 16 $\times$, with 28–110 KB transcripts.
- The verifier’s large-table exponent dropped in two measured steps, $\Theta(x) \rightarrow \Theta(\sqrt{x}) \rightarrow \Theta(x^{1/4})$ (fitted slopes 1.0 $\rightarrow$ 0.50 $\rightarrow$ 0.26; Figure 2).
- Reach: the full compressed chain verified $\pi(2^{26} - 1) = 3\,957\,809$ at $x \approx 6.7 \times 10^7$, with the claimed value matching an independent from-scratch Eratosthenes sieve (prover wall ≈ 3.6 h, allocation-bound).
- Adversarial: self-consistent cheating provers — including ones that corrupt a DP entry and recompute all downstream layers coherently, forge an opening, or tamper a codeword — were rejected in every trial.

[numerical, reproducible from the repository logs]

5.4 The certificate wall, and why it is the same wall

Pushing total verification below $\sqrt{x}$ requires shrinking the certificate. We measured its size at $\alpha = 0.473$ ($\Theta(\sqrt{x})$), dominated by the $K = \pi(\sqrt{x})$ *sequential* outer layer reductions; every batchable (independent) witness is already polylog. Two findings make this a genuine wall, not an artifact:

- **Layering-inherent (construction).** Merging j consecutive layers does not drop the exponent below 0.5 in either a direct or a fully-batched reduction; the cost just moves between transcript, verifier work, and prover fill ($\rightarrow x^{3/2}$ as $j \rightarrow \log K$). A sub- $\sqrt{x}$ certificate cannot come from merging layers.

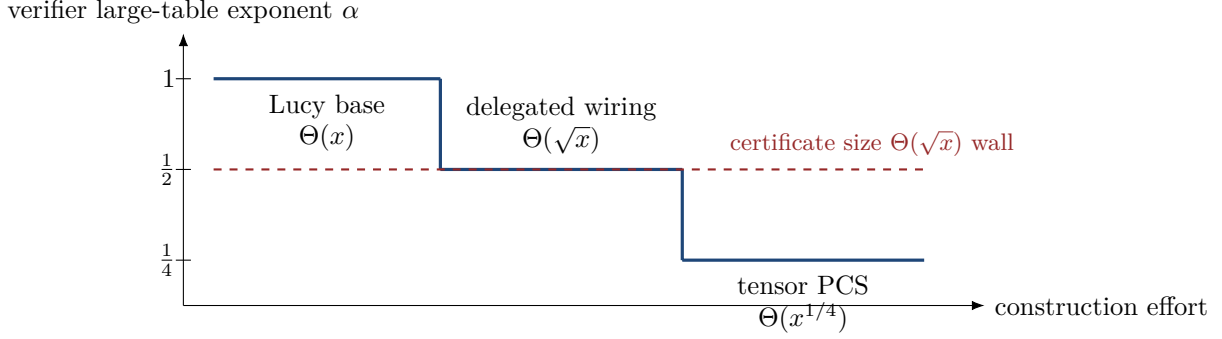


Figure 2: The verifier-exponent staircase. Engineering drives the large-table verifier work down to $\Theta(x^{1/4})$, but the certificate size itself is a $\Theta(\sqrt{x})$ floor (dashed): *total* verification cannot beat $\sqrt{x}$ without compressing the $K = \pi(\sqrt{x})$ sequential layer reductions.

- **Information-forced (lower bound, for this class).** The K sieve checkpoints $\{\varphi(x, a)\}$ are integer-independent: the residual matrix against any polylog smooth predictor is full rank ($\text{rank}/K \rightarrow 0.97$), so their joint information is $\Theta(\sqrt{x}) \cdot \text{polylog}$. Hence the $\sqrt{x}$ certificate is information-forced for *any sieve-reconstructing* verifier. (A single $\pi(x)$ value is still $O(\log x)$ bits; the barrier is joint across layers.) A fundamentally different witness is not ruled out — that routes through the $\#P$ -hardness question of §7.

The certificate-size wall, the verifier-work wall, and the prover-time wall are the same $\lfloor v/p \rfloor$ -semigroup wall on three faces. [numerical; the “information-forced” claim is proven only for the sieve-reconstruction class]

5.5 Honest scope

We are explicit about the limits of this result, because it is easy to oversell.

- **No computation speedup.** The prover pays the full $\tilde{O}(x)$ sieve cost; verification being cheap does not make $\pi(x)$ cheap. This is an adjacent-problem positive: computation unchanged, *trust* exponentially cheaper (a kilobyte-scale transcript replaces months of independent recomputation for record $\pi(x)$ values).
- **Standard tools.** The protocol composes LFKN sum-check [18], GKR-style layer reductions [10], CMT prover techniques [6], and a Liger/Brakedown tensor commitment [19, 5]. The genuinely new ingredients are narrow: the *target* (no prior verification protocol for π), the automaton-MLE treatment of the sieve wiring, and the public-scalar observation.
- **Existence is a corollary.** That a *polylog*-verifier IP exists at all is immediate from GKR over a uniform AKS circuit; we claim no novelty there, only a practical $\tilde{O}(\sqrt{x})$ verifier and the measured walls. An earlier draft’s “polylog or $\sqrt{x}$ floor?” question is resolved: polylog in principle, $\sqrt{x}$ in any sieve-reconstructing construction.
- **Unconditional vs. assumed.** The $\tilde{O}(\sqrt{x})$ verifier is information-theoretically sound; only the $x^{1/4}$ large-table compression assumes a hash/random oracle.

Measure	$\pi(x) \bmod 2$	Random
GF(2) ANF sparsity (nonzero monomials)	0.5000 ($N \leq 20$)	0.50
Approximate degree (ℓ_∞ , $\varepsilon=0.49$)	$\lceil N/2 \rceil$	$N/2$
2-party communication matrix rank	$2^{N/2-1}+2$	$2^{N/2}$
PTF degree (LP-verified)	$N/2$	$N/2$
LFSR / linear complexity	$N/2$	$N/2$
OBDD size	$2^{0.72N}$	$2^{0.80N}$
Entropy rate of $\delta(n)$	5.8 bits	≈ 6
Power spectrum of $\delta(n)$	$1/f^{1.69}$, no lines	no structure
Truth-table gzip ratio (<i>the lone deviation</i>)	2.0–2.6 $\times$	1.0 $\times$

Table 5: Nine of the 36 measures (full list in the source). The only deviation from random, gzip-compressibility, is explained by the trivial density bias ($1/\ln x$ creates long zero-runs) and carries no exploitable circuit structure.

6 Structural measurements

Beyond the negative catalogue, the project produced phenomenology we find genuinely interesting — facts about the prime indicator that are crisp, measured, and (to our knowledge) not previously assembled. We present three. All are [numerical/empirical]; we are explicit about what they do *not* prove.

6.1 The prime parity sequence is pseudorandom under 36 measures

Let $\chi_P(m) = [m \in P]$ and consider $\pi(x) \bmod 2$ as a Boolean function of $N = \log_2 x$ bits (with its smooth-removed companion $f(x) = \pi(x) - R(x)$). The project ran 36 independent complexity measures on it — algebraic, communication, approximation-theoretic, spectral, information-theoretic, and circuit-synthesis — and in *every* case the function lands within the band expected of a random function on N variables. A representative slice is Table 5.

Observation 6.1 (The $N/2$ phase transition). Five measures of different type — approximate degree, communication rank, PTF degree, LFSR length, and the per-bit R -correlation crossover — independently place a sharp threshold at $N/2$. This is the Boolean-complexity shadow of §2: the top $N/2$ bits are the smooth R^{-1} part, the bottom $N/2$ are the GUE-random residual. [numerical, $N \leq 20$]

Remark 6.2 (A hard limit on what this can mean — Natural Proofs). This is strong *indirect* evidence, and it is essential to say what it *cannot* do. Each of the 36 measures is a *natural* property in the Razborov–Rudich sense [24] (efficiently computable on truth tables, satisfied by most functions). Hence *no amount* of such evidence can prove $\pi(x) \bmod 2$ has no small circuits: a successful “natural” proof would break standard cryptographic generators. Pseudorandom is also not the same as hard — cryptographic PRGs pass every efficient test yet have small circuits. And all measurements are at $N \leq 20$. The honest reading: $\pi(x) \bmod 2$ *behaves* like a pseudorandom function, but proving it *is* hard is equivalent to an unconditional circuit lower bound.

6.2 The read-once width spectrum *is* the Hardy–Littlewood census

Reading χ_P as a read-once branching program, let W_j be the (j -th cut) Nerode width and let $D_k(x)$ be the number of distinct aligned length- k windows of χ_P below x ; then $W_j = D_k(2^n)$ with $k = 2^{n-j}$. The project measured the full profile and found a clean two-regime law.

Observation 6.3 (Plateau + admissibility cliff). $W_j = 2^j$ exactly through $j = 16$ (every aligned window of primes is distinct — the global-pseudorandom regime), then collapses. The cliff is exactly the Hardy–Littlewood admissible-tuple census:

$$D_k(x) = A_k + 1,$$

where A_k is the number of admissible aligned patterns of length k and the $+1$ is the single small-prime window at the origin. The cliff depth is $\rho^*(k)$, the maximal admissible weight (3, 5, 9, 16, 28 at $k = 8, 16, 32, 64, 128$) — the functional inverse of the narrowest-admissible- k -tuple diameter $H(k)$ (OEIS A008407, Hensley–Richards [13]). The middle cut is maximal, $W_{n/2} = 2^{n/2}$, exceeding the project’s earlier MLE rank 2050 at $n = 12$, so that factor-2 deficiency is linear-algebraic, not informational. [numerical; $D_k = A_k + 1$ verified to $k = 32$, A_k -entropy law conjectured]

The census saturates as predicted ($k = 8$: $14 = 13+1$; $k = 16$: $107 = 106+1$; $k = 32$ converging), and the growth obeys the conjectural entropy law $A_k = e^{(1+o(1))\pi(k)}$, with the entropy constant $\lim \log_2(A_k)/k$ measured drifting $0.46 \rightarrow 0.37$ over $k = 8 \rightarrow 32$. We could *not* close the $o(1)$: $k = 128$ has $A_k \sim e^{25}$ and defeats direct enumeration, so a transfer-matrix/generating-function count is the well-posed successor. What is new here is the bridge: the *same curve* carries the project’s global “ χ_P is pseudorandom” regime (the 2^j plateau) and its local “ χ_P is Hardy–Littlewood rigid” regime (the cliff). [the bridge is a framing; the pieces are measured/conjectured as tagged]

6.3 Decision-problem geography: a fair coin against the right guess

Finally, the decision problem $[p(n) \leq \hat{x}] \Leftrightarrow [\pi(\hat{x}) \geq n]$ has a measurable “geography” depending on the guess $\hat{x}$.

Observation 6.4 (The better the guess, the fairer the coin). Against the biased guess $\hat{x} = \text{li}^{-1}(n)$, the comparison bit is *constant* (30/30 in the measured range) — the Rubinstein–Sarnak bias [26], predictable up to the Skewes scale $\sim 10^{316}$. Against the centred guess $\hat{x} = R^{-1}(n)$, the bit is an empirical coin flip (10/30 vs 20/30). The closer the smooth guess, the closer the residual bit is to fair — which is the information barrier of §2 restated as a decision problem: all the structure is in the smooth part, and what remains is incompressibly balanced. [numerical]

7 Complexity status of $\mathcal{L}_\pi$

We collect what can be said, conditionally and as framing, about the language

$$\mathcal{L}_\pi = \{(x, c) : \pi(x) = c\}$$

in the binary-input length $n = \log_2 x$. We have *no* unconditional separation; the value of this section is to locate $\mathcal{L}_\pi$ precisely and to record two clean dichotomies. Everything here is [conditional/framing]; the one proven part is the trivial $\#P$ membership.

7.1 Upper bound and the NP-completeness obstruction

Observation 7.1 ($\pi \in \#P$; no short $\#$ -witness). $\pi(x)$ counts the n -bit integers $m \leq x$ satisfying a $\text{poly}(n)$ -time predicate (primality, by AKS [2]), so $\pi \in \#P$, $\mathcal{L}_\pi \in C=P$, and $\{(x, c) : \pi(x) \geq c\} \in PP$. This is folklore and is an upper bound only: the natural $\#$ -witness for the Legendre sieve $\varphi(x, a)$ has 2^a terms, i.e. $2^{\Theta(\sqrt{x}/\log x)}$, doubly exponential in n . [proven (folklore)]

Witness family	leading power α	rung
Enumeration NP-cert (Pratt for each prime; a factor for each composite)	0.985	$\Theta(x)$
Sieve transcript (the protocol of §5)	0.490	$\Theta(\sqrt{x})$
Joint-information floor (the $\pi(\sqrt{x})$ checkpoints)	0.486	$\Theta(\sqrt{x})$
Zeta-zero / explicit formula (Galway $K \sim c\sqrt{x} \log^2 x$)	0.500	$\Theta(\sqrt{x})$
Hypothetical polylog witness (ruled out for sieve class)	-0.000	$\text{poly}(n)$

Table 6: The witness-size ladder (exponents fit as $\log \text{bits} = \alpha \log x + \delta \log \log x + c$, $k = 10\text{--}20$). The three nontrivial natural families agree at $\alpha \approx \frac{1}{2}$ to within 0.014.

Observation 7.2 (NP-completeness would collapse the hierarchy). Since π is a function, a $\text{poly}(n)$ NP-witness for $\pi(x) = c$ immediately yields a coNP -witness for $\pi(x) \neq c$ (exhibit the true $c' = \pi(x)$ with its witness, $c' \neq c$). Hence $\mathcal{L}_\pi \in \text{NP} \Rightarrow \mathcal{L}_\pi \in \text{NP} \cap \text{coNP}$, and $\mathcal{L}_\pi \text{ NP-complete} \Rightarrow \text{NP} = \text{coNP}$. The open question is therefore mere NP-membership — the existence of a $\text{poly}(n)$ certificate — not completeness. [\[proven implication; membership open\]](#)

7.2 The witness-size ladder

How large must a certificate for $\pi(x) = c$ be? Every *natural* family the project could identify sits at one of three rungs (Table 6). Three independent families — the sieve transcript, the analytic/zeta witness, and the joint-information floor — converge at the $\sqrt{x}$ rung; enumeration sits a full power higher; and the polylog rung is ruled out *for the sieve-reconstruction class* by the information-forcing of §5. No natural witness reaches $\text{poly}(n)$.

7.3 Why combinatorial #P-hardness is circular

Observation 7.3 (Parsimonious reduction is the goal in disguise). A parsimonious reduction $\#A \rightarrow \pi$ sends $w \mapsto x(w)$ with $\pi(x(w)) = \#A(w)$. To realise a target count c , the image must lie in $[p_c, p_{c+1} - 1]$, so the map $c \mapsto x(c)$ is the inverse-prime function $p(\cdot)$ — the project’s goal. The “easy direction” of the reduction is therefore as hard as π itself: combinatorial #P-hardness via target-count realisation is *circular*. (The sieve instances are also lattice-structured — sets forced to be multiples of d — with no room to embed an arbitrary #P-complete count.) Consequently, if π is #P-hard at all, the hardness must be a *value-incompressibility* statement, not an instance embedding. [\[framing; precise obstruction, not a non-hardness proof\]](#)

Remark 7.4 (A correction to the project’s own record). An earlier project note (and `CLOSED_PATHS.md` row 175) asserted “exact $\pi(x)$ is #P-hard.” That is an unsubstantiated claim and we retract it: the established fact is $\pi \in \#P$; #P-hardness is open. We flag this because the looser claim, left standing, would have implied false downstream conclusions.

7.4 A conditional Turing-reduction barrier and a dichotomy

Observation 7.5 (SETH forbids natural Turing reductions). Under SETH [14], no polynomial-time Turing reduction $\#\text{SAT} \rightarrow \pi$ can have query-blowup $c < c^* = 1/\alpha$, where α is π ’s time exponent (measured $\alpha \approx 0.66\text{--}0.70$ for the combinatorial π , giving $c^* \approx 1.5$; $\alpha = \frac{1}{2}$ analytically gives $c^* = 2$): otherwise one could simulate π by its own x^α algorithm and solve $\#\text{SAT}$ in $2^{o(n)}$. The natural (parsimonious) blowup is $c \rightarrow 1 < c^*$, so every *natural* Turing reduction is forbidden — the adaptive analogue of Observation 7.3. [\[conditional on SETH\]](#)

Artifact	Computes	Measured behaviour	Exact?
R^{-1} approximator	smooth prediction of $p(n)$	Newton 2–3 iters, scale-free; $\approx 70\text{--}90\%$ of bits	no (smooth)
Lucy/Meissel $\pi(x)$	exact $\pi(x)$	$x^{0.66\text{--}0.70}$; $\approx 0.3\text{--}2$ ms/call (C) to 10^8	yes
Aggarwal $\times$ Dusart $\times$ BPSW	exact $p(n)$	verified to $n = 10^7$ ($p = 179\,424\,673$); $\approx 25\text{--}65\%$ over Aggarwal-only	yes
Sum-check verifier	checks exact $\pi(x)$	$\tilde{O}(\sqrt{x})$ verifier; reach $x \approx 6.7 \times 10^7$	yes (verify)

Table 7: The working artifacts. The $p(n)$ hybrid preserves Aggarwal’s worst-case $O(\sqrt{n} \log^4 n)$ and improves only the practical constant; its one regularity is that the optimal narrowing knob obeys $K^* \sim \sqrt{\text{bracket width}}$ in the fast- π regime.

Corollary 7.6 (The exclusion dichotomy). *Assuming $P \neq NP$, π **computable in polylog time and π being $\#P$ -hard are mutually exclusive**: a polylog algorithm would put a $\#P$ -hard problem in (uniform) $\text{poly}(n)$ time, collapsing the hierarchy. Equivalently, a $\#P$ -hardness proof for π would prove the project’s goal impossible. The two faces of the problem — find the algorithm, or prove hardness — are the two ways the question can resolve, and exactly one can hold.* [conditional ($P \neq NP$); framing]

On present evidence $\mathcal{L}_\pi$ is an NP-intermediate-flavoured counting problem with a $\Theta(\sqrt{x})$ certificate and no proven $\text{poly}(n)$ one — precisely the status the $\tilde{O}(\sqrt{x})$ verifier of §5 realises from above and the information floor pins from below.

8 What actually runs

To keep the paper grounded, we list the working, tested artifacts behind the claims above (Table 7). Each is a real program with a selftest and a results file; none is a polylog exact algorithm, and we do not present any as a contribution to the asymptotic state of the art.

The honest summary of this table is short: we can compute the *smooth half* of $p(n)$ in polylog time, we can compute $\pi(x)$ exactly only in time $x^{2/3}$ -ish, and we can *verify* an exact $\pi(x)$ in $\tilde{O}(\sqrt{x})$. The polylog target touches none of these.

9 Open problems and where a breakthrough must come from

9.1 The central gap

The one sentence worth repeating: the gap between the proven lower bound $\Omega(\log x)$ and the proven upper bound $O(x^{1/2+\varepsilon})$ — from $\text{poly}(n)$ to $2^{n/2}$ — is essentially unstudied for a natural problem, and closing it in either direction is the whole question. By Observation 4.1 this is the same as asking whether $\pi(x) \in \text{NC}$ (indeed whether it has $\text{poly}(n)$ -size circuits), and by Corollary 7.6 a hardness resolution would go through $\#P$ -hardness.

9.2 Four concrete frontiers

The project leaves four sharply-posed sub-questions, each of which is a clean target in its own right:

1. **Growing-dimension matrix powering in TC^0 .** Fixed-dimension MPOW is in TC^0 , and $\mathbb{F}_{2^n}$ exponentiation is in TC^0 via Frobenius [12], but the AKS route to $\text{PRIMES} \in \text{TC}^0$ needs *growing-dimension* ($k = \text{polylog}$) powering over $\mathbb{Z}_n$, which sits exactly at the TC^0/NC^1 boundary and is genuinely open.
2. **#P-hardness of π .** The sole remaining lower-bound lever (Observation 7.3): if true it must be a value-incompressibility statement, and by Corollary 7.6 it would settle the headline question negatively.
3. **GapL / determinantal form.** Is there a $\text{poly}(n)$ -size, logspace-uniform matrix $A(x)$ with $\det A(x) = \pi(x)$ (i.e. $\pi \in \text{GapL}$)? This is the sharpest positive target (Observation 4.2); it would bypass the workspace mismatch entirely.
4. **The sequential-sieve / layer-batching wall.** Algebraically compress the semigroup generated by $v \mapsto \lfloor v/p \rfloor$. A sub- $\sqrt{x}$ certificate (even computational) would be new and would crack the $\Theta(\sqrt{x})$ wall of §5; merging layers provably does not suffice.

A smaller, self-contained number-theory target also remains: closing the $o(1)$ in the census entropy law $A_k = e^{(1+o(1))\pi(k)}$ at $k = 128$ (a transfer-matrix or generating-function count), which the direct enumeration cannot reach.

9.3 What a breakthrough would have to be

The localisation of §2 says the positive direction requires *computing a $\Theta(\sqrt{x})$ -fold cancellation in an oscillatory sum with GUE-random phases without forming the summands* — a genuine closed form where none is known. The negative direction requires *super-polynomial uniform circuit lower bounds for an explicit function*, which must clear the Natural Proofs barrier (Remark 6.2). Both are at or beyond the current frontier of their fields. It is entirely possible that the problem requires mathematics that does not yet exist; nothing we found rules out a fast algorithm, and nothing we found produces one.

10 Methodology and honesty statement

10.1 How the work was produced

This material is the synthesis of a long-running *autonomous* research loop: independent cycles, each of which read a single live state document, performed one piece of falsification-first work, and recorded the outcome before stopping. Three disciplines shaped the corpus and are worth stating because they affect how the results should be read:

- **Catalogue-first.** Before any approach was pursued, it was checked against the closed-paths catalogue; closing an approach meant adding a row with its mechanism. This is why the negative results are unusually complete and why the taxonomy of §3 has no leftover bin.
- **Code or it didn’t happen.** Each experiment is a single CLI-parameterised script with a `-selftest` (boundary cases become permanent regression tests) and a results file stating *what would falsify the result*. The numbers in this paper are drawn from those files and logs.
- **Honest reporting as a hard constraint.** Failures are filed as failures with the mechanism; earlier overclaims (including the project’s own — the “information-theoretic barrier” phrasing of Remark 2.3, the “ π is #P-hard” assertion we retracted, the “polylog or $\sqrt{x}$ ” question we resolved) are corrected when found wrong.

10.2 Limitations

We list these plainly so the reader can calibrate.

- **No external peer review.** The corpus has not been refereed; this manuscript is the first attempt to expose it to outside scrutiny.
- **Small N .** Many structural measurements are at $N \leq 20$ ($x \leq 10^6$) and the verifier reach is $x \approx 6.7 \times 10^7$; asymptotic claims extrapolate scaling fits and should be read as such.
- **Synthesis-level “theorems.”** The four obstruction theorems (§4) are arguments, not refereed proofs; their rigorous cores are standard facts specialised to π .
- **Conjectural cores.** The GUE law (and everything resting on it) is numerical/conjectural; the $\#P$ and SETH statements are conditional or framing; the census entropy law is conjectured.
- **Novelty bar is internal.** “Novel” in the project sense means “not in the catalogue and not obvious from one careful reading of the literature”; we do not claim priority against the entire literature, and the verification protocol explicitly composes known tools.

11 Conclusion

Computing $\pi(x)$ or $p(n)$ exactly in time $\text{polylog}(x)$ remains open, and this paper does not move the headline bounds. What it offers instead is a map. The hardness is localised to the residual $\delta(n) = p(n) - R^{-1}(n)$: small in information ($O(\log x)$ bits), expensive in computation ($\Theta(\sqrt{x})$ apparent), and encoding the GUE-random phases of $\sqrt{x}$ zeta zeros — a *computational*, not information-theoretic, barrier. Over 700 catalogued attacks fail in one of three ways, driven by a short list of mechanisms that all reduce to “you must aggregate $\Theta(\sqrt{x})$ quantities.” The one direction that escapes is *verification*: exact $\pi(x)$ is succinctly checkable, with a working $\tilde{O}(\sqrt{x})$ verifier and a $\Theta(\sqrt{x})$ certificate wall we show to be information-forced for the sieve class. And the phenomenology is sharp: the prime parity is pseudorandom under 36 measures, its read-once width spectrum *is* the Hardy–Littlewood census, and the decision problem is a fair coin against the right smooth guess. The problem sits exactly at the frontier — a fast algorithm would place a pseudorandom-looking, $\#P$ function in NC, while proving impossibility would require circuit lower bounds beyond current technique. We expect progress to require new mathematics, and we have tried to say precisely which mathematics, and precisely how confident one should be in each step along the way.

A The closed-path taxonomy

Table 8 reproduces the family inventory of `status/CLOSED_PATHS.md` with the dominant failure mode of each. Aggregate mode tallies (from the catalogue’s own summary, ≈ 641 classified): circularity $C \approx 82$, equivalence $E \approx 222$, information loss $I \approx 180$; the remainder are *partial* (≈ 19), *works-but-slow* (≈ 8), *exists but useless* (≈ 3), or *open* (5). As noted in §3, the catalogue’s headline count drifts across revisions (545+/641+/730+); we use “over 700.”

B Selected verified edges

The project maintains an internal catalogue of “edges” (verified mathematical facts), each graded by an internal evidence-strength tag (EVS: H/M/L/shape). Table 9 lists a representative selection cited above; the grades are the project’s own and are not peer-reviewed.

Family (catalogue heading)	Dominant mode	Typical mechanism
Analytic / explicit formula / zeta zeros	E, I	M1, M2, M4
Algebraic / number theory / p -adic	C, E	M1, M6
Sieve / combinatorial / counting	E, I	M2, M7
Machine learning / statistical / fitting	I	M5
Quantum computing / quantum information	E, I	M2, M9
Dynamical systems / automata	I	M9
Topological / exotic mathematics	E	M1
Information theory / complexity theory	E, I	M2, M5, M9
Encoding / novel representations	C, I	M2, M6
Additive combinatorics / Goldbach / partitions	E, I	M2, M5
Gap prediction / interpolation	I	M5
Spectral methods	E	M1
Communication / streaming / preprocessing / boosting	E, I	M2, M5

Table 8: Catalogue families and their dominant failure mode/mechanism. Mechanisms M1–M9 are defined in §3.

ID	Statement	EVS
E1.1	$E(x) = \pi(x) - \text{li}(x)$ carries only $O(\log x)$ bits.	H
E1.3	Per-bit difficulty of $p(n)$ is a sharp ≈ 4 -bit sigmoid.	H
E1.4	$N/2$ universality (degree/rank/PTF/LFSR/correlation crossover).	M
E2.1	MPS bond-dimension identity at every primordial cut (wheel- W gain $\phi(W)/W$).	H
E2.2	Liouville/parity identity $\pi(x) = \frac{x-L(x)}{2} - C_3(x)$.	M
E3.3	Riemann’s $R(x)$ recovers $\approx 50\%$ of the digits, exactly.	H
E3.4	Spectral truncation gives a unique prime in a tiny window.	H
E4.4	Telescoped (Meissel–Lehmer) parallel depth $= \pi(x^{1/3})$.	L
E5.1	BPSW correctness $\Rightarrow \text{PRIMES} \in \text{TC}^0$.	H
E5.3	Growing-dimension MPOW in TC^0 — the only open frontier.	H
E6.6	Aggarwal 2025 binary-search optimality for $p(n)$.	H
E6.8	Dusart bracketing: p_n in an interval of width $\sim n$.	M
E7.1	Zeta zeros are GUE-typical / “independent” to order 6.	shape

Table 9: Selected edges from EDGES.md. EVS is the project’s internal evidence grade, not an external rating.

C Reproducibility and file map

Every numerical claim traces to a script (with `-selftest`) and a `*_results.md` in the repository. The principal locations are:

- `proven/` — barriers (`barriers.md`, `complexity.md`, `information.md`) and the four obstruction notes (`circuit_size_barrier.md`, `uniformity_barrier.md`, `workspace_mismatch_barrier.md`, `convergence_acceleration_barrier.md`).
- `novel/` — `succinct_verification_of_pi.md`, `failure_taxonomy.md`, `info_computation_gap.md`, `pseudorandomness_of_pi.md`.
- `status/CLOSED_PATHS.md` — the full closed-path catalogue; `EDGES.md` — the verified-edge catalogue.
- `experiments/constructions/p12_sumcheck_pi_verification/` — the verification stack (`compressed_layer.py`, `leaf_open.py`, `pcs_commit.py`, `sharpP_probe.py`,

turing_reduction_barrier.py, reproduction logs).

- `experiments/analytic/` — convergence, zero-scaling, sparse-precision, Wasserstein, guess-comparison-oracle, and local-pattern-census experiments.
- `algorithms/aggarwal_dusart_bpsw/` — the $p(n)$ hybrid.

D Headline constants

Quantity	Measured value
R -inverse Newton iterations (scale-free)	2–3
Free-bit fraction k_0/B (declining in x)	$0.88 \rightarrow 0.71$
Width of $\delta(n)$	$\sim x^{0.488}$
Entropy rate of $\delta(n)$	5.8 bits/value
Time-bounded description length $K^t(\delta_{1..N})$	$\approx 5.58 N$
Power spectrum of $\delta(n)$	$1/f^{1.70}$
Min. zeros for exact $\pi(x)$	$N_{\min} \sim x^{0.49}$
Verifier large-table exponent (staircase)	$1.0 \rightarrow 0.50 \rightarrow 0.26$
Certificate-size exponent	$0.473 (\Theta(\sqrt{x}))$
Verification reach	$x = 2^{26} - 1 \approx 6.7 \times 10^7$
Pseudorandomness measures passed	36
Census law	$D_k(x) = A_k + 1, A_k = e^{(1+o(1))\pi(k)}$
Census entropy constant $\log_2(A_k)/k$	$0.46 \rightarrow 0.37 (k = 8 \rightarrow 32)$
$p(n)$ hybrid practical gain over Aggarwal-only	$\approx 25\text{--}65\%$

Table 10: The paper’s headline measured constants, with sources in Appendix C.

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